

An engineered probiotic for the inhibition of *Salmonella* via tetrathionate-induced production of microcin H47

Jacob D Palmer, Emma Piattelli, Beth McCormick, Mark Silby, Christopher Bringham, and Vanni Bucci

ACS Infect. Dis., Just Accepted Manuscript • DOI: 10.1021/acsinfecdis.7b00114 • Publication Date (Web): 17 Sep 2017

Downloaded from <http://pubs.acs.org> on September 19, 2017

Just Accepted

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a free service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are accessible to all readers and citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.



1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

An engineered probiotic for the inhibition of *Salmonella* via tetrathionate-induced production of microcin H47

Jacob D. Palmer¹, Emma Piattelli², Beth A. McCormick^{3,4}, Mark W. Silby^{2,5}, Christopher J. Brigham^{1,5}, Vanni Bucci^{2,4,5*}

¹Department of Bioengineering, University of Massachusetts Dartmouth, 285 Old Westport Rd, N. Dartmouth, MA 02747

²Department of Biology, University of Massachusetts Dartmouth, 285 Old Westport Rd, N. Dartmouth, MA 02747

³MaPS, University of Massachusetts Medical School, 55 Lake Avenue North Worcester, MA 01655

⁴UMass Center for Microbiome Research, 55 Lake Avenue North Worcester, MA 01655

⁵UMassD Probiotic Discovery, Engineering and Manufacturing Pipeline, 285 Old Westport Rd, N. Dartmouth MA 02747

*Correspondence to: Vanni Bucci, vbucci@umassd.edu

KEYWORDS: Microbiome, Engineering, Probiotics, *Salmonella*, Tetrathionate, Microcins, *Escherichia coli* Nissle

ABSTRACT: Complications arising from antibiotic-resistant bacteria are becoming one of the key issues in modern medicine. Members of drug-resistant *Enterobacteriaceae* spp. include opportunistic pathogens (e.g. *Salmonella* spp.) that are among the leading causes of morbidity and mortality worldwide. Overgrowth of these bacteria is considered a hallmark of intestinal dysbiosis. Microcins (small antimicrobial peptides) produced by some gut commensals can potentially cure these conditions by inhibiting these pathogens, and have been proposed as a viable alternative to antibiotic treatment. In this proof-of-concept work, we leverage this idea to develop a genetically engineered prototype probiotic to inhibit *Salmonella* spp. Upon exposure to tetrathionate, a molecule produced in the inflamed gut during the course of *Salmonella* infection. We developed a plasmid-based system capable of conferring the ability to detect and utilize tetrathionate, while at the same time producing microcin H47. We transferred this plasmid-based system to *Escherichia coli* and demonstrated the ability of the engineered strain to inhibit growth of *Salmonella* in anaerobic conditions while in the presence of tetrathionate, with no detectable inhibition in the absence of tetrathionate. In direct competition assays between the engineered *E. coli* and *Salmonella*, the engineered *E. coli* had a considerable increase in fitness advantage in the presence of 1mM tetrathionate as compared to the absence of tetrathionate. In this work, we have demonstrated the ability to engineer a strain of *E. coli* capable of using an environmental signal indicative of intestinal inflammation as an inducing molecule, resulting in production of a microcin capable of inhibiting the organism responsible for the inflammation.

Medical complications related to the emergence of antibiotic-resistant bacteria is a major issue in modern healthcare due to the resulting increase in morbidity, mortality, length of hospitalization and related healthcare costs¹. The CDC estimates that every year, more than 2 million people acquire multi-drug-resistant infections, resulting in over 23,000 related deaths². *Enterobacteriaceae* spp. include opportunistic pathogens (Carbapenem-resistant *Klebsiella* spp., Fluoroquinolone-resistant *Salmonella* spp., adherent-invasive *Escherichia coli*) that are among the leading causes of morbidity and mortality worldwide^{3,4}. According to the CDC, drug-resistant *Salmonella* spp. are responsible for more than 100,000 infections, while their non-resistant counterparts already account for 1.2 million infections and 450 deaths in the US every year². It was recently demonstrated that microcin production by specific gut bacteria can ameliorate these conditions by inhibiting certain pathogens⁵. However, in these organisms, microcin production is dependent upon a narrow range of environmental conditions^{6,7} and regulatory mechanisms which are, in many cases, not fully understood⁵.

Engineered microbes as diagnostic tools or therapeutics have recently been proposed as “smart” alternatives to traditional drugs. For the former, recent remarkable examples have included engineering microbes to sense cancer or inflammation deriving from the production of reactive oxygen species by the host^{8,9}. For the latter, major examples have included the construction of non-pathogenic *E. coli* that kill *Pseudomonas aeruginosa* by means of microcins or pyocins, elicited through this pathogen’s quorum-sensing signaling system^{10–12}. A recent report from Sassone-Corsi et al., demonstrated that the production of microcins from *E. coli* strain Nissle 1917 (*EcN*) can therapeutically displace pathogenic *Enterobacteriaceae* species in specific environmental conditions (e.g. iron limitation)⁵. Our group independently verified that the class IIb sideromycin, microcin H47 (MccH47)¹³, a high molecular mass, post-translationally modified peptide, originally isolated from *E. coli* strain H47, can inhibit *Salmonella* growth *in vitro*. Therefore, rather than relying on the wild-type regulatory mechanisms, we sought to develop a probiotic with a specifically designed induction system for MccH47.

During gut inflammation, reactive oxygen species produced by the host react with luminal thiosulfate, resulting in production of tetrathionate¹⁴. *Salmonella* species utilize the gene products of the *ttr* operon (*ttrRSBCA*), which provide this pathogen with the ability to utilize tetrathionate as a terminal electron acceptor, conferring a growth advantage over the competing microbiota during inflammation conditions¹⁴. The *ttrBCA* genes of *Salmonella*, encoding the three subunits of tetrathionate reductase, have previously been transferred to *E. coli*, resulting in functional tetrathionate reductase activity¹⁴. The *ttrBCA* promoter is positively regulated by TtrR in the presence of tetrathionate, and by Fnr, under anoxic conditions¹⁴. Very recently, the *ttrRS* genes of *Salmonella*, encoding the tetrathionate sensor and response regulator, were chromosomally integrated into *E. coli* strain NGF-1 and served as a functional tetrathionate detection probiotic in a mouse model of inflammation for over 6 months⁹. In light of these advances, we developed an engineered strain of *EcN* harboring a plasmid-based system carrying *mchAXIBCDEF* and *ttrRSBCA*, capable of producing MccH47 in response to environmental tetrathionate, resulting in the ability to inhibit and out-compete *Salmonella* during *in vitro* experiments.

Results and Discussion

Construction and analysis of plasmid-based system for the inducible production of microcin H47

Plasmid pJPMcH47 was developed for the L-rhamnose dependent production of MccH47, and constructed by Gibson Assembly (see Methods). pJPMcH47 contains all *mch* genes of *E. coli* H47 (*mchAXIBCDEF*), with the *mchXIB* genes immediately downstream of the *rhaP_{BAD}* promoter (Fig. 1A). The aim of this design is to specifically regulate production of the MccH47-precursor (MchB), based on L-rhamnose concentration. For the purpose of *in vitro* assays, we developed *E. coli* strain NEB10β pJPMcH47 and *EcN* pJPMcH47, and then assessed the ability of each strain to inhibit *Salmonella enterica* subsp. *Enterica* serovar Typhimurium (hereafter referred to as *S. Typhimurium*) and *E. coli* strain DH5α.

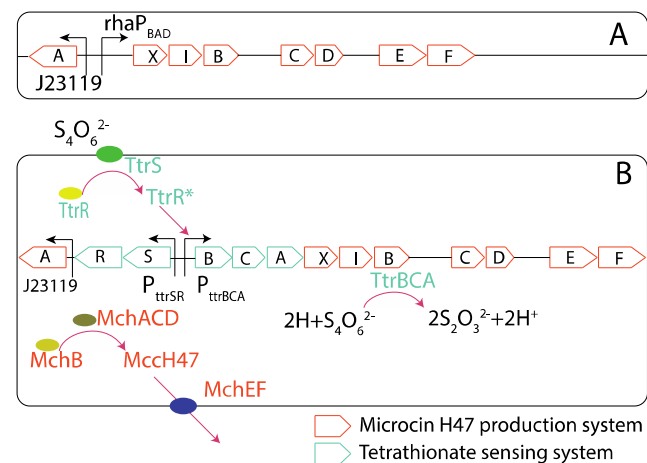


Figure 1: (A) Plasmid pJPMcH47 for the L-rhamnose-induced production of MccH47. (B) Plasmid pptrMcH47 for the tetrathionate-induced production of MccH47. Orange indicates genes and reactions needed for MccH47 production, light blue indicates genes and reaction needed from tetrathionate sensing.

Inhibition assays were based on previous work with microcin J25 by Delgado et al.¹⁶, where inhibition was evaluated visually by measuring a zone of inhibition in susceptible lawns grown on agar plates previously stabbed with a strain carrying

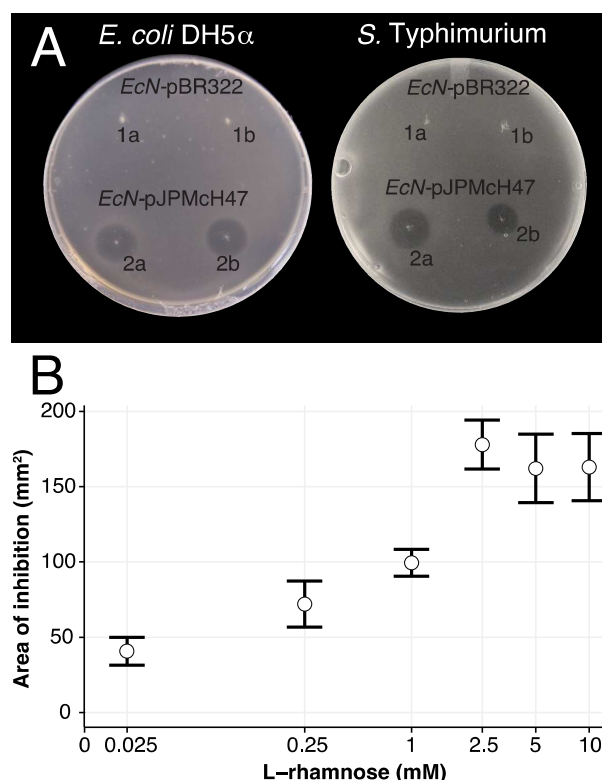


Figure 2: (A) Inhibition assays comparing the effect of *EcN* WT and *EcN* pJPMcH47 against lawns of pathogenic *S. Typhimurium* (left) and *E. coli* DH5α (right) on minimal media, supplemented with 22mM L-rhamnose and 0.2mM 2,2'-dipyridyl (see Methods). (B) Measurement of inhibition area in *S. Typhimurium* lawn as a function of L-rhamnose concentration on LB agar plates supplemented with variable L-rhamnose (0.025mM-10mM) and 0.2mM 2,2'-dipyridyl. 2,2'-dipyridyl addition was necessary for maximum *S. Typhimurium* inhibition by MccH47 (see Methods and Fig. S2).

pJPMcH47 and inactivated with chloroform and UV. Both *EcN* pJPMcH47 (Fig. 2A) and *E. coli* NEB10β pJPMcH47 (Fig. S1) were capable of inhibiting both *S. Typhimurium* and *E. coli* DH5α. As expected, based on the nature of the L-rhamnose dependent induction system utilized¹⁷, inhibition increases with increasing L-rhamnose concentration, while *EcN* WT showed no inhibition (Fig. 2B). Taken together, these results demonstrate that the MccH47 produced by our engineered strains is the causative agent in the inhibition of *S. Typhimurium*. This is in line with reported literature, showing that MccH47 produced from *EcN* WT is unable to inhibit *S. Typhimurium* even during iron starvation^{5,6}, while MccH47 produced from our donor *E. coli* strain H47 is able to inhibit *S. Typhimurium*^{6,13}. One reason could be the lack of post-translational modifications in MccH47 derived from *EcN* WT¹⁸, potentially due to the lack of *mchA*, a necessary gene for mature MccH47 antibacterial activity. Another possibility could be the observed divergence in amino acid sequence for genes *mchE* and *mchF* between *EcN* and *E. coli* H47⁶.

Construction and analysis of plasmid-based tetrathionate-detection system

Plasmid pptrMcH47 (Fig. 1B) was developed to confer utilization of tetrathionate reductase activity and tetrathionate dependent production of McchH47, and was constructed by Gibson Assembly (see Methods). Plasmid pptrMcH47 contains all genes of the *ttr* operon from *S. Typhimurium* (*ttrRSBCA*), and all genes necessary for mature McchH47 production, immunity, and secretion (*mchAXIBCDEF*). *mchXIB* is encoded immediately downstream of *ttrA* resulting in co-transcription along with *ttrBCA* from the *ttrBCA* promoter. To assess successful tetrathionate-induced McchH47 production, we compared the inhibitory effect of *EcN* pptrMcH47 against *E. coli* DH5 α in variable conditions (Fig. 3). Again, we observed that McchH47-dependent inhibition increases in iron limited conditions (Fig 3A-B). This is consistent with previous studies that have shown that species susceptible to inhibition by McchH47 express catecholate siderophore receptors *ironJ/fui*, *cir*, *fepA*⁶, which serve as the mode of entry into susceptible cells. Importantly, this should not constitute an issue with respect to the functioning of our construct *in vivo*, as it is known that the iron limitation is a typical feature of a host's inflamed gut¹⁸. When testing for tetrathionate addition in iron-limitation, we observe the expected emergence of inhibition only in the presence of tetrathionate (Fig. 4A). Supplementation to the media with 0.1% D-glucose reduced inhibition across all experimental conditions (Fig 3A,C), which is noteworthy in light of published reports which utilize glucose supplementation alongside P_{ttrRS}-based induction and McchH47 production^{9,19}.

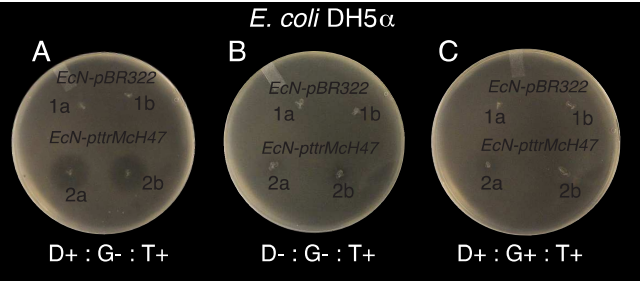


Figure 3: Characterization of McchH47-based inhibition of *E. coli* DH5 α by *EcN* pptrMcH47, with *EcN* pBR322 control, as a function of 0.2mM 2,2'-dipyridyl (D), 0.1% D-Glucose (G) and 1 mM of potassium tetrathionate (T). + stands for compound present, - stands for compound absent.

Analysis of tetrathionate-induced inhibition

We proceeded to test the tetrathionate-induced inhibition *in vitro* against *S. Typhimurium*. First, solid media inhibition assays were carried out anaerobically in LB agar, again supplemented with 0.2mM 2,2'-dipyridyl, with and without 1.5 mM potassium tetrathionate. In media containing potassium tetrathionate, *EcN* pptrMcH47 was capable of inhibiting *S. Typhimurium* while *EcN* WT did not (inhibition zone not detected) (Fig. 4A) confirming the ability of our construct to inhibit *S. Typhimurium* at a physiologically relevant disease-state¹⁴.

While static plate assays demonstrate the functional capability of the constructs, these experiments do not account for the effect of competition for growth nutrients. Therefore, we performed competition experiments between *EcN* pptrMcH47 and either *S. Typhimurium* WT or *S. Typhimurium* pBR322. As-

says were conducted anaerobically, in LB broth supplemented with 0.2mM 2,2'-dipyridyl, in presence or absence of 1mM potassium tetrathionate, and supplemented with 100 μ g/mL carbenicillin when both strains contained resistance. In experiments to analyze growth rate, we determined 1 mM of potassium tetrathionate to be a plausible concentration resulting in no significant fitness cost due to tetrathionate addition (Fig. S3). Competition experiments were initiated with a *S. Typhimurium* (pBR322) - *EcN* pptrMcH47 ratio of approximately 1:1 (Fig. 4B). Bacteria were grown for a total of 10 hours, and 10-fold dilutions were plated onto MacConkey agar for colony enumeration. For both strains, we followed Lenski et al.²⁰ and our previous work^{21,22} to determine relative fitness as

$$W_i = \frac{\Delta P_i}{P_i(0)} = \frac{P_i(end) - P_i(0)}{P_i(0)} \quad (1)$$

where $P_i(t)$ is the proportion of strain *i* at time *t*. We performed linear regression analysis and fitted the model $W_{SE} \sim 1 + Tet + Tag$ where *Tet* and *Tag* are two “dummy” variables, with *Tet* indicating absence/presence of tetrathionate and *Tag* indicating use of *S. Typhimurium* WT or *S. Typhimurium* pBR322. Results of linear regression show a significant decrease in *S. Typhimurium* fitness at 1 mM potassium tetrathionate irrespective of *S. Typhimurium* WT or *S. Typhimurium* pBR322 ($p < 0.02$). We also observe that the inhibitory effect is reduced when competition is performed against *S. Typhimurium* WT in a media lacking antibiotic addition ($p < 0.05$). However, because we do not detect significant differences in growth curve dynamics in isolation for any of the strains used in this experiment, neither in presence nor in absence of tetrathionate (Fig. S3), we speculate that this reduced competitiveness against *S. Typhimurium* WT may be due to plasmid loss in the absence of antibiotic. When testing for an increase in *S. Typhimurium* proportion observed at 600 minutes compared to 360 minutes for the *EcN* pptrMcH47 vs *S. Typhimurium* WT, we do not observe statistical significance ($p > 0.05$) between the two data points. We acknowledge, but cannot explain the recovery by *S. Typhimurium* pBR322 from 360 minutes to 600 minutes. Our results confirm the ability of *EcN* pptrMcH47 to suppress *S. Typhimurium* growth beyond what is obtained in direct competition experiments in an environment without the supplementation of tetrathionate (Fig. 4C). The increased competitive advantage of *EcN* pptrMcH47 over *S. Typhimurium* in an environment supplemented with tetrathionate is particularly important considering recent work which has indicated that tetrathionate in the lumen of the inflamed gut provides a growth advantage for *S. Typhimurium* over the rest of the competing microbiota⁴.

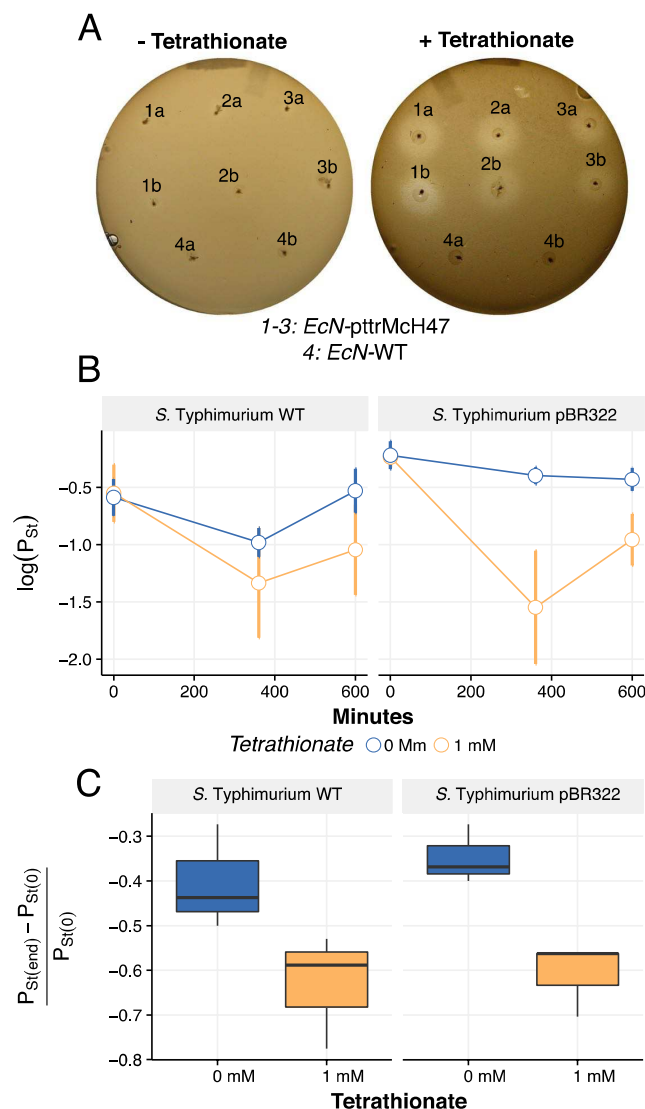


Figure 4: Tetrathionate-induced inhibition by *EcN* ptrMcH47 McH47 production. (A) Inhibition assays comparing the ability of three separate clones of *EcN* ptrMcH47 and *EcN* WT to inhibit growth *S. Typhimurium* in absence (left) and presence (right) of 1mM potassium tetrathionate (a,b indicate duplicate stabs). (B) Results from *in vitro* ecological competition experiments, showing the proportion of *S. Typhimurium* WT (left) or *S. Typhimurium* pBR322 (right) over time. (C) *S. Typhimurium* fitness estimation obtained from competition experiments (see²¹). Supplementation of 1mM potassium tetrathionate leads to production of McH47 and a statistically significant reduction ($p < 0.05$) in *S. Typhimurium* fitness.

Outlook

The use of microcins for the treatment of enterobacterial colitis can be regarded as a novel and potent alternative to antibiotics²³. Here we have developed a genetically engineered *E. coli* that produces a *S. Typhimurium*-inhibiting microcin in response to tetrathionate, a molecule resulting from the inflammation of the intestine, a hallmark of a *Salmonella* infection⁴. Using *in vitro* assays, we have shown that *EcN* ptrMcH47 not only prevents *S. Typhimurium* growth in static agar inhibition assays, but also significantly reduces *S.*

Typhimurium fitness in pairwise ecological competition experiments.

While current focus on engineered probiotics (with respect to gut pathogen infections) has been mostly directed towards disease sensing and reporting^{8,9}, our work provides a first proof-of-concept towards using a genetically engineered living therapeutic to induce a specific microbiome correction during the course, or at the onset of, a particular disease state.

The realization that microcins play a crucial role in gut microbiome ecology⁵ opens a new door for the treatment and prevention of a range of gut pathogens. As new microcins or related antimicrobial peptides with narrow target specificity are further discovered and characterized²⁴, our work provides a demonstration of a plausible delivery method for timely gut pathogen eradication with minimal off-target effects, as compared to broad spectrum antibiotics.

To obtain a proof-of-concept system that would ensure a fully functional inflammation sensing–pathogen killing phenotype, without the need of any extra strain-specific chromosomal features, we included all the necessary genes for tetrathionate sensing and McH47 production into a single DNA construct. We then transferred this system to *EcN*, an FDA-approved probiotic that has been shown amenable towards engineered probiotic applications^{8,11}. For illustrative purpose, and in line with recent studies on the construction and preliminary testing of engineered bacteria^{8,10,11,25}, we opted for a plasmid-based approach. However, for translation to clinics, as well as for trials using *in vivo* models, future work will be performed to achieve stable chromosomal integration. This may involve using a background genome with promising characteristics regarding host colonization and retention of recombinant genes, such as *E. coli* NGF-1⁹.

A possible issue that is common to biocontrol through supplementation of native and engineered bacteria comes from the emergence of mutations that may alter the regulation or function of the system. However, in light of recent work that demonstrated wildtype stability and mutational rate of the *ttr* operon in an engineered *E. coli*⁹, we expect our probiotic to properly work over the time frame of a *Salmonella* infection. Moreover, future efforts to properly control for this possible issue, could lead to the development of a probiotic that autolyzes within a certain time frame after supplementation²⁶. As such, utilizing this method, it is possible to envision a single probiotic bacterium capable of colonizing a host, sensing a disease state or particular pathogen, and eradicating the disease-causing organism without the need for physician intervention or use of broad-spectrum antibiotics.

Methods

Microbial Strains, Media, and Growth Conditions

Strains used in this study include *Escherichia coli* strain NEB10 β (New England Biolabs, Ipswich, MA), *E. coli* strain DH5 α (New England Biolabs, Ipswich, MA), *E. coli* strain Nissle 1917 (gift from)²¹ and/or *Salmonella enterica* subsp. enterica; serovar Typhimurium ATCC 29630 (purchased from ATCC, location). Plasmid constructs developed in this work were first transformed by electroporation into *E. coli* NEB10 β cells, and then to *E. coli* Nissle (*EcN*). Oligonucleotides used in this study are listed in Table 1. All media and additional reagents listed in this study were purchased from Sigma Aldrich, St. Louis, MO, unless otherwise indicated.

pJPMcH47 and ptrMcH47 were constructed using standard methods for Gibson Assembly²⁵, and the Gibson Assembly

Master Mix (New England Biolabs, Ipswich, MA). To construct pJPMcH47, a linear version of pBR322 was produced by polymerase chain reaction using primers pBR322FWD and pBR322REV, and select regions of pLJV3 (encoding *rho*_{BAD}, *CFP*, *mccF*, and *mchA*) and pEX2000 (encoding *mchX-IBCDEF*) were amplified using primer sets pLJV3FWD/pLJV3REV and pEX2000FWD/pEX2000REV, respectively. pptrMcH47 was constructed by amplification of pJPMcH47 using primers pJPMcH47FWD/pJPMcH47REV and amplification of the *ttrRSBCA* operon of *S. Typhimurium* using primers SentFWD/SentREV. *ttrRSBCA* was included in the plasmid as opposed to just *ttrRS* because in Riglar et al.⁹ the constructed sensor strain, when in direct competition with *Salmonella*, only detected tetrathionate if the *Salmonella* was incapable of tetrathionate utilization (Δ ttrR).

Solid Media Inhibition Assays

Inhibition assays in solid media were designed and carried out based on previous work by Delgado et al.¹⁶. Briefly, *E. coli* strains were grown overnight on LB agar plates, supplemented with carbenicillin (100µg/mL) when relevant. Individual colonies were then used to inoculate 3mL of LB broth, supplemented with carbenicillin (100µg/mL) when relevant. 1µL of liquid culture was then used to create an agar stab in solid media and incubated at 37°C, either aerobically or anaerobically, for 24 hours. Post incubation, cells were inactivated with chloroform and UV. Molten 3% agar was then added to an overnight culture of susceptible cells to a final concentration of 0.75%, and then 3mL of the mixture was overlaid on top of the inactivated agar stab plates and allowed to solidify. Plates were then incubated aerobically overnight at 37°C and inhibition halos were observed. When inhibition halo size was a relevant characteristic, ImageJ software (<https://imagej.nih.gov/ij/>) was utilized. For solid medium inhibition assays of *S. Typhimurium* and *E. coli* DH5α by *EcN* pJPMcH47, agar stabs were made in M9 minimal salts supplemented with 0.1mM CaCl₂, 2mM MgSO₄, 0.2mM 2,2'-dipyridyl, and 0.4% L-rhamnose. For variable L-rhamnose concentration inhibition experiments, culture stabs of *EcN* pJPMcH47 were made in LB agar supplemented with 0.2mM 2,2'-dipyridyl and L-rhamnose, ranging from 0.25µM to 10mM. All aspects of inhibition assays utilizing plasmid pJPMcH47 were performed aerobically.

Table 1. Strains, Plasmids, and Oligonucleotides

	Relevant characteristics/sequence (5' - 3')	Source or Reference
Strains		
<i>Escherichia coli</i> Nissle 1917		27
<i>Salmonella</i> Typhimurium	<i>Salmonella enterica</i> subsp. enterica serovar Typhimurium ATCC 29630	ATCC
<i>Escherichia coli</i> NEB10β	Δ (ara-leu) 7697 araD139 fhuA Δ lacX74 galK16 galE15 e14- ϕ 80dlacZ Δ M15 recA1 relA1 endA1 nupG rpsL (StrR) rph spoT1 Δ (mrr-hsdRMS-mcrBC)	New England Biolabs
<i>Escherichia coli</i> DH5α	F- Φ 80lacZ Δ M15 Δ (lacZYA-argF) U169 recA1 endA1 hsdR17 (rK-, mK+) phoA supE44 λ - thi-1 gyrA96 relA1	New England Biolabs

For solid medium inhibition assays of *S. Typhimurium* by *EcN* pptrMcH47, culture stabs in LB agar supplemented with 0.2mM 2,2'-dipyridyl and 1.0 or 1.5 mM (see Main Text) potassium tetrathionate were incubated anaerobically at 37°C for 24 hours in Oxoid anaerobic jars with anaerobic atmosphere generation bags. Notably, potassium tetrathionate was not added to the LB agar medium until the temperature had reached 50°C, and was prepared immediately before introduction to the media and sterilized by filtration using 0.22µm filter membrane. Upon removal from the jars, cells were immediately inactivated, overlaid with *S. Typhimurium* culture in 0.75% agar, and incubated aerobically overnight. Because microcin H47 mode of action relies on functional catecholate siderophore receptors *iroN*, *cir*, *fepA*⁶, 2,2'-dipyridyl addition was essential to render *S. Typhimurium* susceptible to inhibition by our probiotic. Control experiments using the laboratory strain *E. coli* DH5a allowed us to rule out necessity of iron limitation for the MccH47 production by our constructs (Fig. 3A,B).

Liquid Media Competition Assays

Competition assays were carried out in triplicate for each experimental condition in a Forma Scientific Model 1025 anaerobic chamber. LB was allowed to equilibrate in anaerobic conditions overnight, prior to initiation of any competition assays. 3mL LB was distributed to individual test tubes, with final concentration of 0.2mM 2,2'-dipyridyl, and both 1mM potassium tetrathionate and 100µg/mL carbenicillin, as indicated. Potassium tetrathionate was prepared immediately before inoculation and sterilized by filtration using 0.22µm filter membrane. Individual colonies of *EcN* pptrMcH47 and *S. Typhimurium* and *S. Typhimurium* pBR322 were taken from LB agar plates, and incubated aerobically overnight at 37°C in 3mL of LB, supplemented with 100µg/mL carbenicillin for plasmid retention when relevant. Liquid cultures were then transferred to the anaerobic chamber, and ~10⁵ cells of each culture were transferred into media for experimental conditions for analysis of relative fitness in competition, based on presence of tetrathionate. Competition assays were set up to compete *EcN* pptrMcH47 with *S. Typhimurium*, and *EcN* pptrMcH47 with *S. Typhimurium* pBR322, in which case 100µg/mL carbenicillin was supplemented to the media.

Plasmids		
pBR322	pMB1, ApR, TcR	New England Biolabs
pLJV3	pUC57, CFP, mchA, rhaPBAD	This study; synthesized by General Biosystems
pJPMcH47	rhaPBAD, mchAXIBCODEF	This study
pttrMcH47	ttrRSBCA, mchAXIBCODEF	This study
pEX2000	pBR322, mchAXIBCODEF	²⁹
Oligonucleotides		
pBR322FWD	GGATTATTTCTATTAAAAATGAGGCCCTTTTCGTCTTCAAGAATTCT	
pBR322REV	GAGATAGCGGTAGCTAACTAGACGTCAGGTGGCACTTTTCG	
pLJV3FWD	TTACCAGACCTCACCCAGACTTCATTACGACCAGTCTAAAAAGCGCCTG	
pLJV3REV	GAAAAGTGCCACCTGACGTCTAGTTAGCTACCGCTATCTCCAACGTGC	
pEX2000FWD	TTTAGACTGGTCGTAATGAAGTCTGGGTGAGGTCTGGTAAGA	
pEX2000REV	CTTGAAGACGAAAGGGCCTCATTTTAAATAGAAATAATCCTGTCAACAGTTCTCAACG	
pJPMcH47FWD	AAAAATCGAGCGTATATAACGTCTGGGTGAGGTCTGGTAAGA	
pJPMcH47REV	TCTGTTGGTTTGATCTGGCGGGATGTGACGATCGTTGACAGC	
SentFWD	TTACCAGACCTCACCCAGACGTTATATACGCTCGATTTTGGCCGC	
SentREV	TGTCAACGATCGTCACATCCCGCCAGATCAAACCAACAGAA	

AUTHOR INFORMATION

Corresponding Author

*Correspondence should be addressed to: Dr. Vanni Bucci, Department of Biology, SENG 335A, 285 Old Westport Road, N. Dartmouth, MA 02445, telephone: +1(508)999-8219, email: vbucci@umassd.edu

ORCID

Vanni Bucci: 0000-0002-3257-2922

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publication website at DOI. Figure S1, *E. coli* Nissle 1917 pBR322 and *E. coli* NEB10 β pJPMcH47 inhibition of *E. coli* DH5 α and *S. Typhimurium*; Figure S2, *E. coli* Nissle 1917 pBR322 and *E. coli* Nissle 1917 pJPMcH47 inhibition of *E. coli* DH5 α and *S. Typhimurium* in presence and absence of 0.2mM 2,2'-dipyridyl; Figure S3, Growth curves in LB media under anaerobic conditions with 0.2mM 2,2'-dipyridyl.

ACKNOWLEDGMENT

VB acknowledges support from the National Institute of Allergy and Infectious Disease (Grant R15-AI12985-01A1), and the National Science Foundation (Grant 1458347). VB, MWS, CJB and BAM acknowledge support from the UMass President Science and Technology Office. BAM acknowledges support from NIH DK056754. JDP acknowledges support through a UMassD Distinguished Doctoral Fellowship. The authors acknowledge Aunnesha Bhowmick, Lindsey Ly and Shakti Bhattacharai for help with experiments and constructive discussion.

REFERENCES

- (1) Medina, E., and Pieper, D. H. (2016) Tackling Threats and Future Problems of Multidrug-Resistant Bacteria, in *How to Overcome the Antibiotic Crisis* (Stadler, M., and Dersch, P., Eds.), pp 3–33. Springer International Publishing. http://dx.doi.org/10.1007/82_2016_492
- (2) Antibiotic Resistance Threats in the United States, 2013 | Antibiotic/Antimicrobial Resistance | CDC.
- (3) Ng, K. M., Ferreyra, J. A., Higginbottom, S. K., Lynch, J. B., Kashyap, P. C., Gopinath, S., Naidu, N., Choudhury, B., Weimer, B. C., Monack, D. M., and Sonnenburg, J. L. (2013) Microbiota-liberated host sugars facilitate post-antibiotic expansion of enteric pathogens. *Nature* 502, 96–99. <http://dx.doi.org/10.1038/nature12503>
- (4) Winter, S. E., Winter, M. G., Xavier, M. N., Thiennimitr, P., Poon, V., Keestra, A. M., Laughlin, R. C., Gomez, G., Wu, J., Lawhon, S. D., Popova, I. E., Parikh, S. J., Adams, L. G., Tsolis, R. M., Stewart, V. J., and Bäuml, A. J. (2013) Host-derived nitrate boosts growth of *E. coli* in the inflamed gut. *Science* 339, 708–711. <http://dx.doi.org/10.1126/science.1232467>
- (5) Sassone-Corsi, M., Nuccio, S.-P., Liu, H., Hernandez, D., Vu, C. T., Takahashi, A. A., Edwards, R. A., and Raffatellu, M. (2016) Microcins mediate competition among Enterobacteriaceae in the inflamed gut. *Nature* 540, 280–283. <http://dx.doi.org/10.1038/nature20557>
- (6) Patzer, S. I., Baquero, M. R., Bravo, D., Moreno, F., and Hantke, K. (2003) The colicin G, H and X determinants encode

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

microcins M and H47, which might utilize the catecholate siderophore receptors FepA, Cir, Fiu and IroN. *Microbiol. Read. Engl.* 149, 2557–2570. <http://dx.doi.org/10.1099/mic.0.26396-0>

(7) Behrens, H. M., Six, A., Walker, D., and Kleanthous, C. (2017) The therapeutic potential of bacteriocins as protein antibiotics. *Emerg. Top. Life Sci.* 1, 65–74. <http://dx.doi.org/10.1042/ETLS20160016>

(8) Daeflfer, K. N.-M., Galley, J. D., Sheth, R. U., Ortiz-Velez, L. C., Bibb, C. O., Shroyer, N. F., Britton, R. A., and Tabor, J. J. (2017) Engineering bacterial thiosulfate and tetrathionate sensors for detecting gut inflammation. *Mol. Syst. Biol.* 13, 923. <http://dx.doi.org/10.15252/msb.20167416>

(9) Riglar, D. T., Giessen, T. W., Baym, M., Kerns, S. J., Niederhuber, M. J., Bronson, R. T., Kotula, J. W., Gerber, G. K., Way, J. C., and Silver, P. A. (2017) Engineered bacteria can function in the mammalian gut long-term as live diagnostics of inflammation. *Nat. Biotechnol.* 35, 7. <http://dx.doi.org/10.1038/nbt.3879>

(10) Saeidi, N., Wong, C. K., Lo, T.-M., Nguyen, H. X., Ling, H., Leong, S. S. J., Poh, C. L., and Chang, M. W. (2011) Engineering microbes to sense and eradicate *Pseudomonas aeruginosa*, a human pathogen. *Mol. Syst. Biol.* 7, 521. <http://dx.doi.org/10.1038/msb.2011.55>

(11) Hwang, I. Y., Koh, E., Wong, A., March, J. C., Bentley, W. E., Lee, Y. S., and Chang, M. W. (2017) Engineered probiotic *Escherichia coli* can eliminate and prevent *Pseudomonas aeruginosa* gut infection in animal models. *Nat. Commun.* 8. <http://dx.doi.org/10.1038/ncomms15028>

(12) Gupta, S., Bram, E. E., and Weiss, R. (2013) Genetically Programmable Pathogen Sense and Destroy. *ACS Synth. Biol.* 2, 715–723. <http://dx.doi.org/10.1021/sb4000417>

(13) Laviña, M., Gaggero, C., and Moreno, F. (1990) Microcin H47, a chromosome-encoded microcin antibiotic of *Escherichia coli*. *J. Bacteriol.* 172, 6585–6588. <http://dx.doi.org/10.1128/jb.172.11.6585-6588.1990>

(14) Winter, S. E., Thiennimitr, P., Winter, M. G., Butler, B. P., Huseby, D. L., Crawford, R. W., Russell, J. M., Bevins, C. L., Adams, L. G., Tsois, R. M., Roth, J. R., and Bäuml, A. J. (2010) Gut inflammation provides a respiratory electron acceptor for *Salmonella*. *Nature* 467, 426–429. <http://dx.doi.org/10.1038/nature09415>

(15) Hensel, M., Hinsley, A. P., Nikolaus, T., Sawers, G., and Berks, B. C. (1999) The genetic basis of tetrathionate respiration in *Salmonella typhimurium*. *Mol. Microbiol.* 32, 275–287. <http://dx.doi.org/10.1046/j.1365-2958.1999.01345.x>

(16) Delgado, M. A., Vincent, P. A., Fariás, R. N., and Salomón, R. A. (2005) YojI of *Escherichia coli* functions as a microcin J25 efflux pump. *J. Bacteriol.* 187, 3465–3470. <http://dx.doi.org/10.1128/JB.187.10.3465-3470.2005>

(17) Haldimann, A., Daniels, L. L., and Wanner, B. L. (1998) Use of new methods for construction of tightly regulated arabinose and rhamnose promoter fusions in studies of the *Escherichia coli* phosphate regulon. *J. Bacteriol.* 180, 1277–1286.

(18) Vassiliadis, G., Destoumieux-Garzón, D., Lombard, C., Rebuffat, S., and Peduzzi, J. (2010) Isolation and Characterization of Two Members of the Siderophore-Microcin Family, Microcins M and H47. *Antimicrob. Agents Chemother.* 54, 288–297. <http://dx.doi.org/10.1128/AAC.00744-09>

(19) Azpiroz, M. F., Rodríguez, E., and Laviña, M. (2001) The Structure, Function, and Origin of the Microcin H47 ATP-Binding Cassette Exporter Indicate Its Relatedness to That of Colicin V. *Antimicrob. Agents Chemother.* 45, 969–972. <http://dx.doi.org/10.1128/AAC.45.3.969-972.2001>

(20) Lenski, R. E., Rose, M. R., Simpson, S. C., and Tadler, S. C. (1991) Long-Term Experimental Evolution in *Escherichia coli*. I. Adaptation and Divergence During 2,000 Generations. *Am. Nat.* 138, 1315–1341. <http://dx.doi.org/10.1086/285289>

(21) Bucci, V., Nadell, and Xavier, J. B. (2011) The Evolution of Bacteriocin Production in Bacterial Biofilms. *Am. Nat.* 178, E162–E173. <http://dx.doi.org/10.1086/662668>

(22) Nadell, C. D., Bucci, V., Drescher, K., Levin, S. A., Bassler, B. L., and Xavier, J. B. (2013) Cutting through the complexity of cell collectives. *Proc. R. Soc. B Biol. Sci.* 280. <http://dx.doi.org/10.1098/rspb.2012.2770>

(23) Cotter, P. D., Ross, R. P., and Hill, C. (2013) Bacteriocins — a viable alternative to antibiotics? *Nat. Rev. Microbiol.* 11, 95–105. <http://dx.doi.org/10.1038/nrmicro2937>

(24) Zschüttig, A., Zimmermann, K., Blom, J., Goesmann, A., Pöhlmann, C., and Gunzer, F. (2012) Identification and Characterization of Microcin S, a New Antibacterial Peptide Produced by Probiotic *Escherichia coli* G3/10. *PLOS ONE* 7, e33351. <http://dx.doi.org/10.1371/journal.pone.0033351>

(25) Holowko, M. B., Wang, H., Jayaraman, P., and Poh, C. L. (2016) Biosensing *Vibrio cholerae* with Genetically Engineered *Escherichia coli*. *ACS Synth. Biol.* 5, 1275–1283. <http://dx.doi.org/10.1021/acssynbio.6b00079>

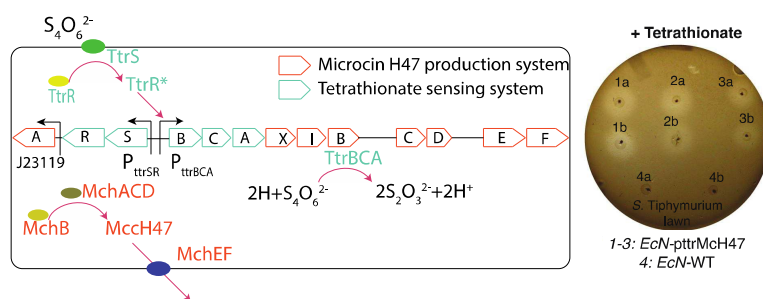
(26) Camacho, E. M., Mesa-Pereira, B., Medina, C., Flores, A., and Santero, E. (2016) Engineering *Salmonella* as intracellular factory for effective killing of tumour cells. *Sci. Rep.* 6, srep30591. <http://dx.doi.org/10.1038/srep30591>

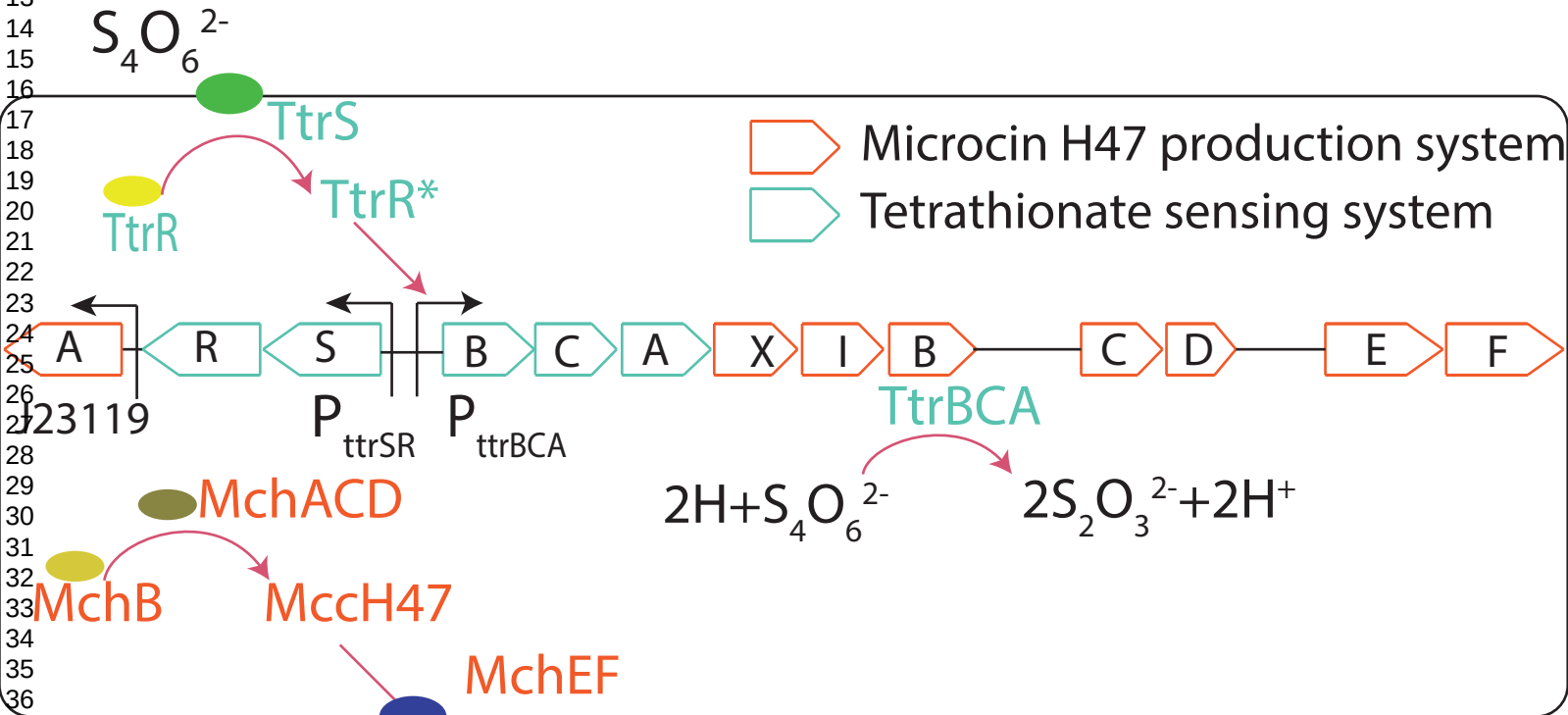
(27) Bucci, V., Tzen, B., Li, N., Simmons, M., Tanoue, T., Børgart, E., Deng, L., Yeliseyev, V., Delaney, M. L., Liu, Q., Olle, B., Stein, R. R., Honda, K., Bry, L., and Gerber, G. K. (2016) MDSINE: Microbial Dynamical Systems INference Engine for microbiome time-series analyses. *Genome Biol.* 17, 1–17. <http://dx.doi.org/10.1186/s13059-016-0980-6>

(28) Gibson, D. G., Young, L., Chuang, R.-Y., Venter, J. C., Hutchison, C. A., and Smith, H. O. (2009) Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nat. Methods* 6, 343–345. <http://dx.doi.org/10.1038/nmeth.1318>

(29) Gaggero, C., Moreno, F., and Laviña, M. (1993) Genetic analysis of microcin H47 antibiotic system. *J. Bacteriol.* 175, 5420–5427. <http://dx.doi.org/10.1128/jb.175.17.5420-5427.1993>

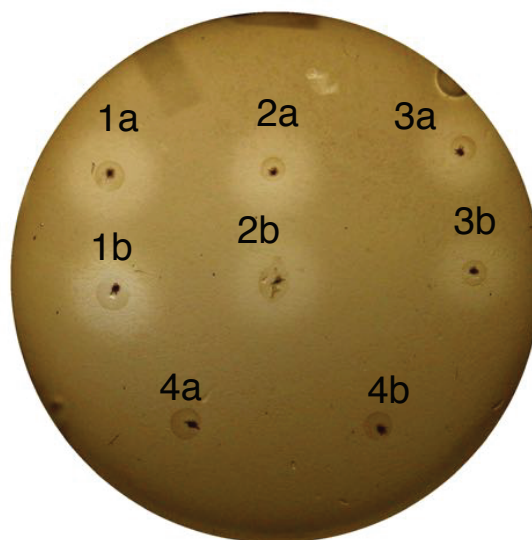
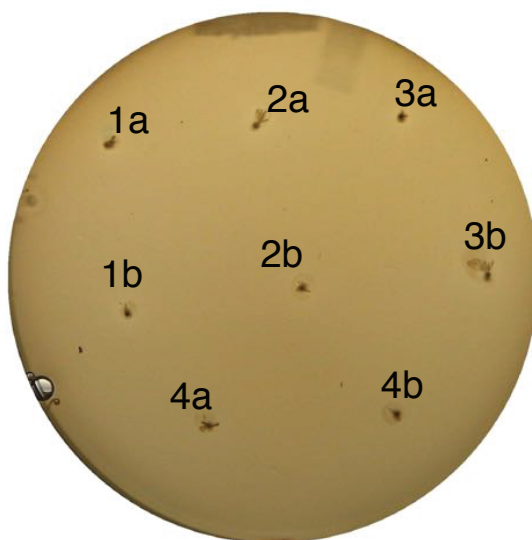
For Table of Contents Use Only





- Tetrathionate

+ Tetrathionate



1-3: *EcN*-pttrMchH47
 ACS Paragon Plus Environment
 4: *EcN*-WT